Robotics in Modern Classrooms

Using LEGO EV3 to Develop 21st-Century Skills in Kenyan Schools

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Abstract

This paper explores how LEGO EV3 robotics can strengthen STEM learning in Kenyan schools by developing computational thinking, collaboration, creativity, communication, and problem-solving skills.

Introduction

Education systems increasingly require learners to develop practical skills alongside academic knowledge. Robotics provides a hands-on environment where students can design, build, test, and improve solutions to real-world problems.

Why Robotics Matters

Robotics transforms learners from passive recipients of information into active creators. LEGO EV3 enables students to connect science, mathematics, engineering, and programming through practical experiences.

Skills Developed Through LEGO EV3

Key skills include computational thinking, teamwork, creativity, communication, resilience, and project management.

Challenges and Recommendations

Schools should start small, invest in teacher development, integrate robotics into curriculum goals, and seek partnerships for sustainability.

Conclusion

Robotics education can help prepare learners for a technology-driven future while strengthening creativity, critical thinking, and innovation.